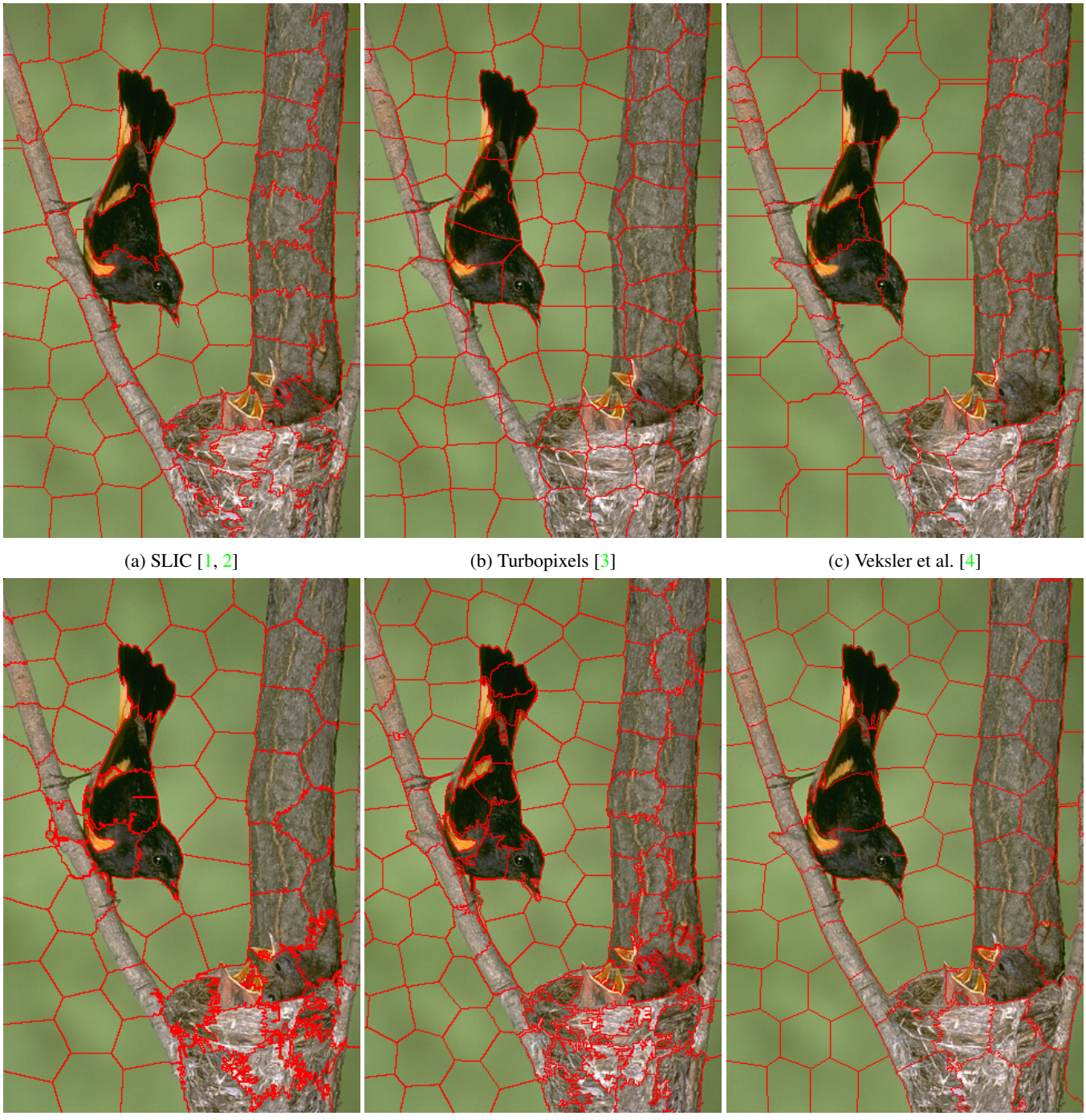
A FAST METHOD FOR INFERRING HIGH-QUALITY SIMPLY-CONNECTED SUPERPIXELS (ADDITIONAL VISUAL-COMPARISON EXAMPLES)

Oren Freifeld, Yixin Li, and John W. Fisher III

Massachusetts Institute of Technology
Computer Science and Artificial Intelligence Laboratory

ABSTRACT

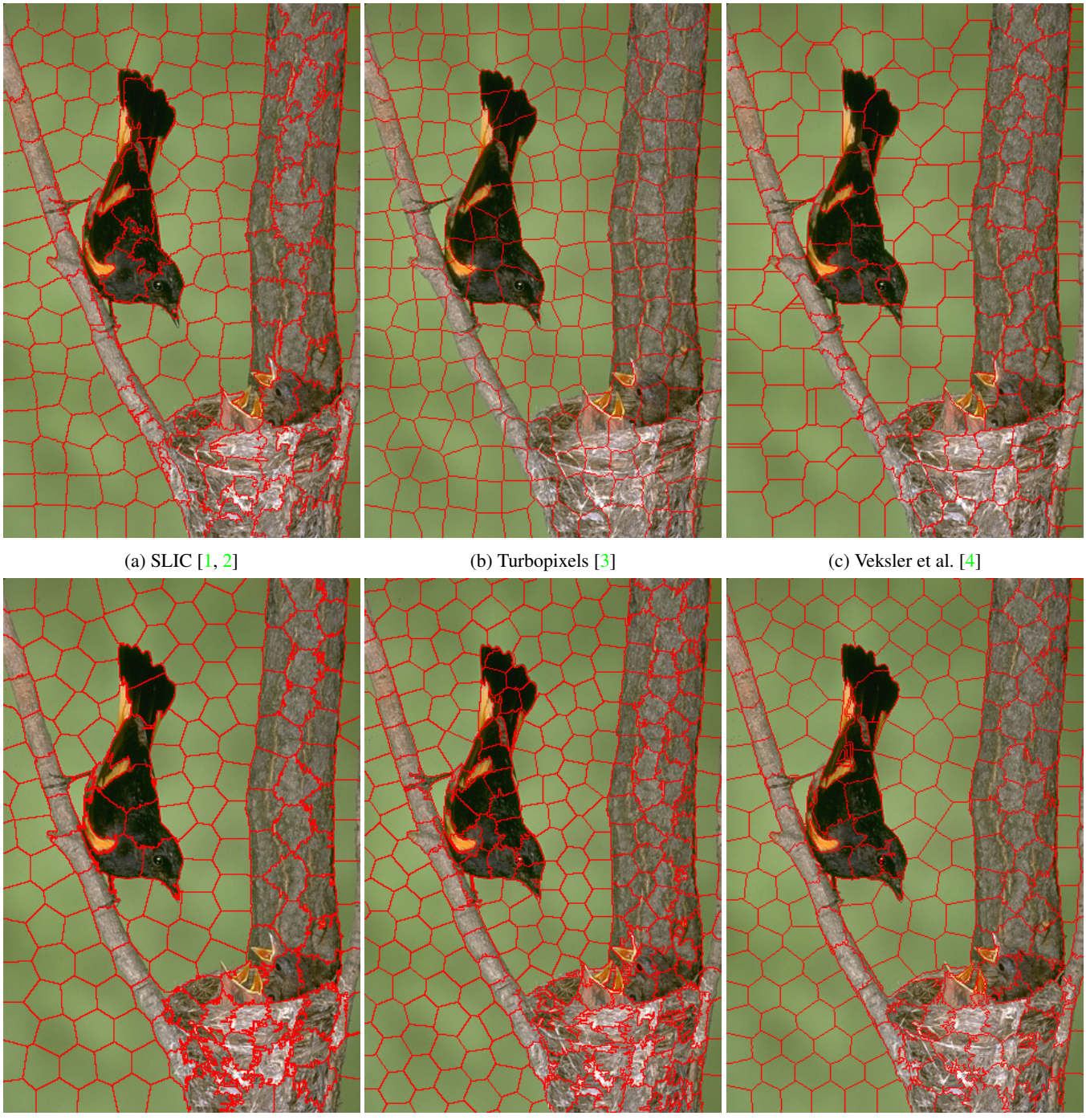
This document is complementary to our ICIP 2015 paper titled "A Fast Method for Inferring High-Quality Simply-Connected Superpixels". It contains additional examples for a visual comparison between the different methods.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

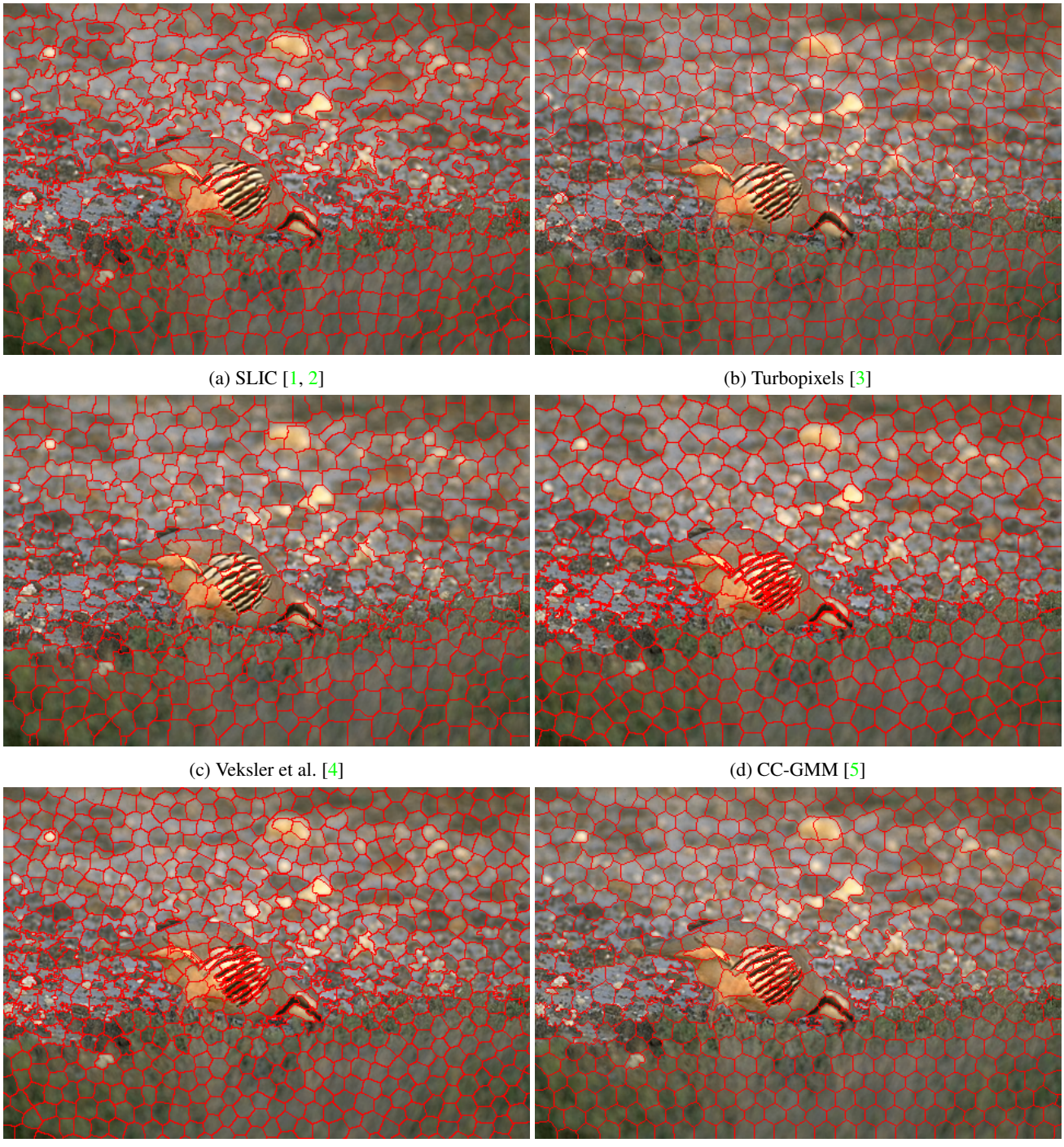
Fig. 1: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 88$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 2: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

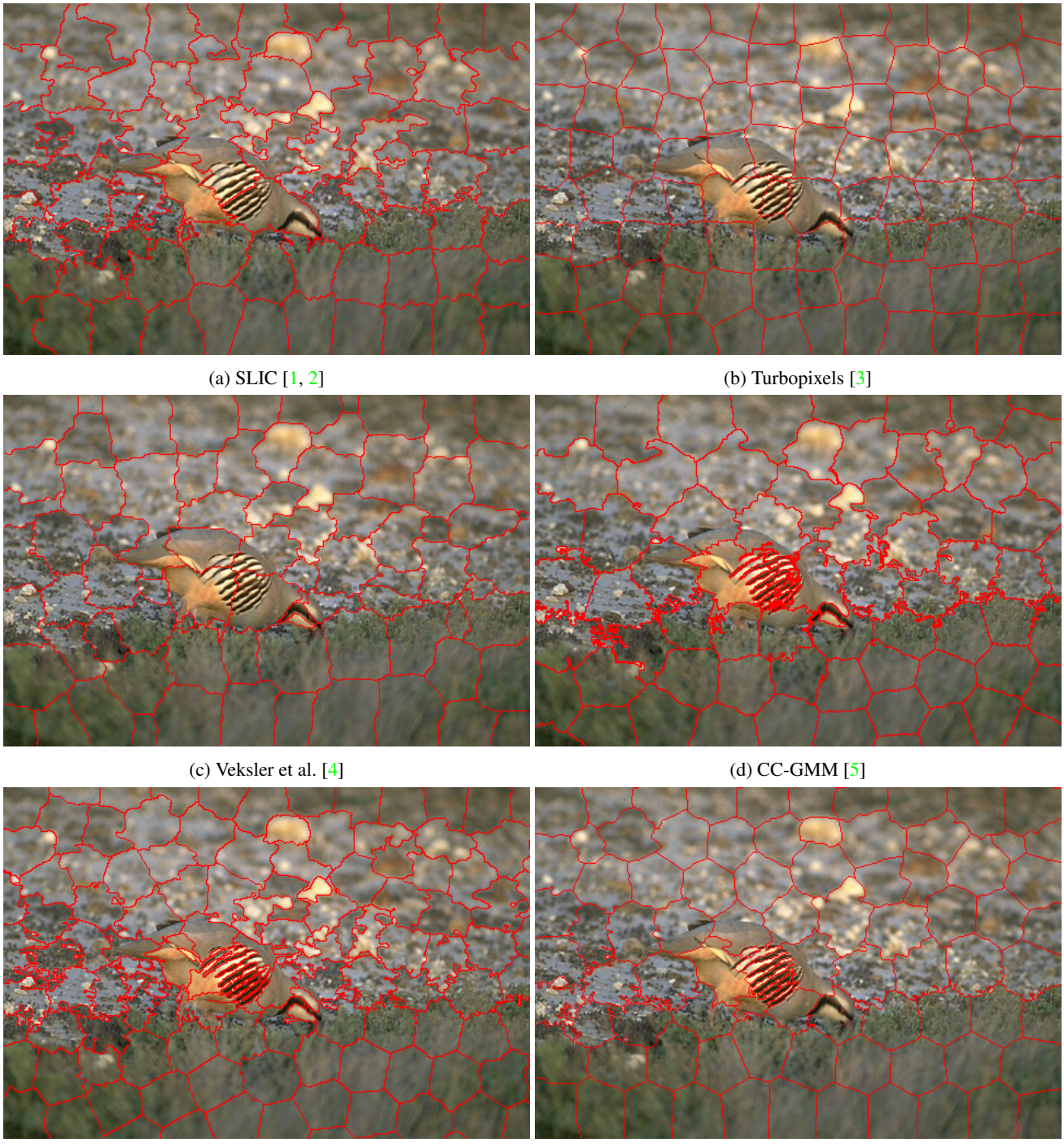
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 3: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

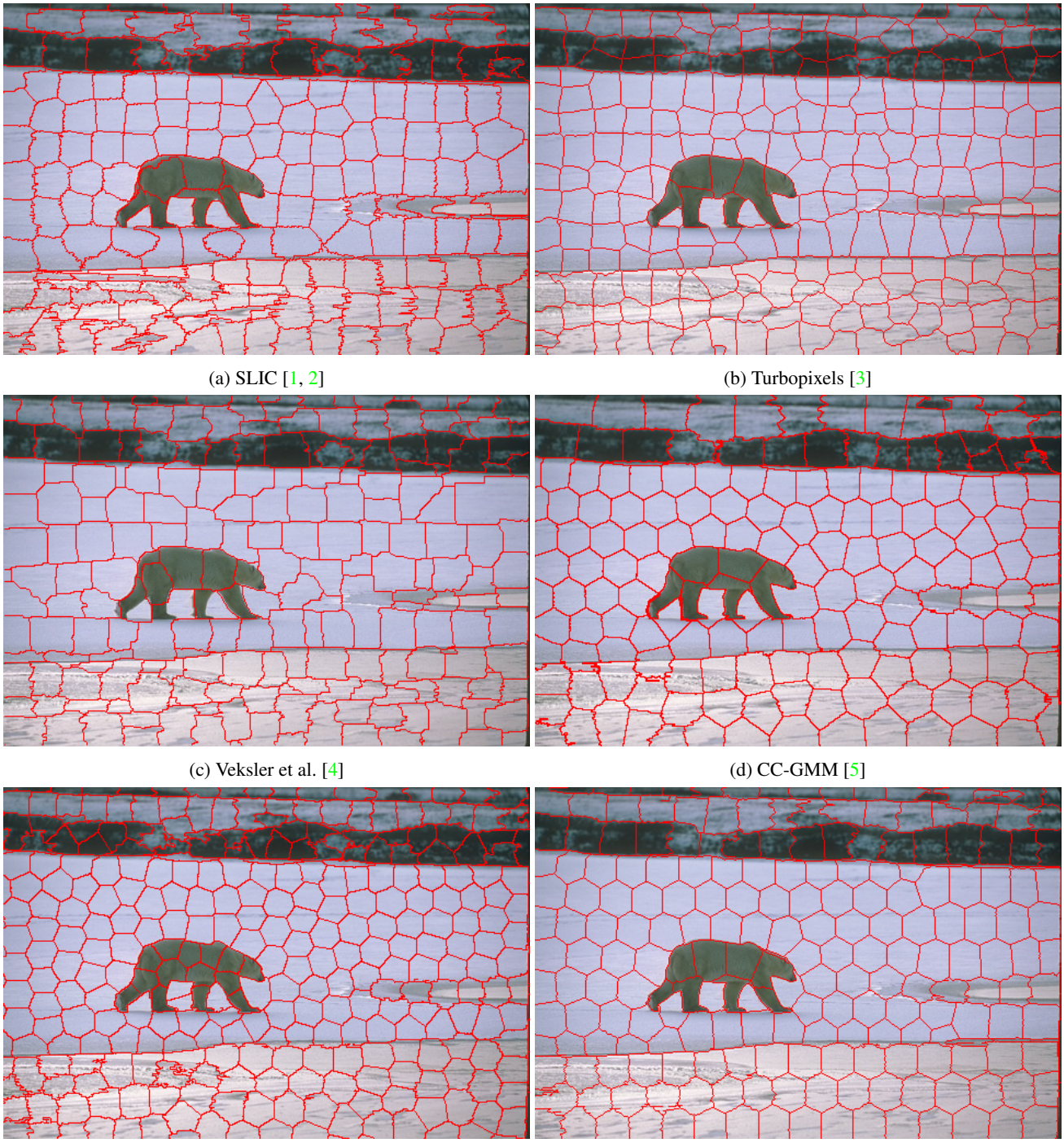
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 4: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 88$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

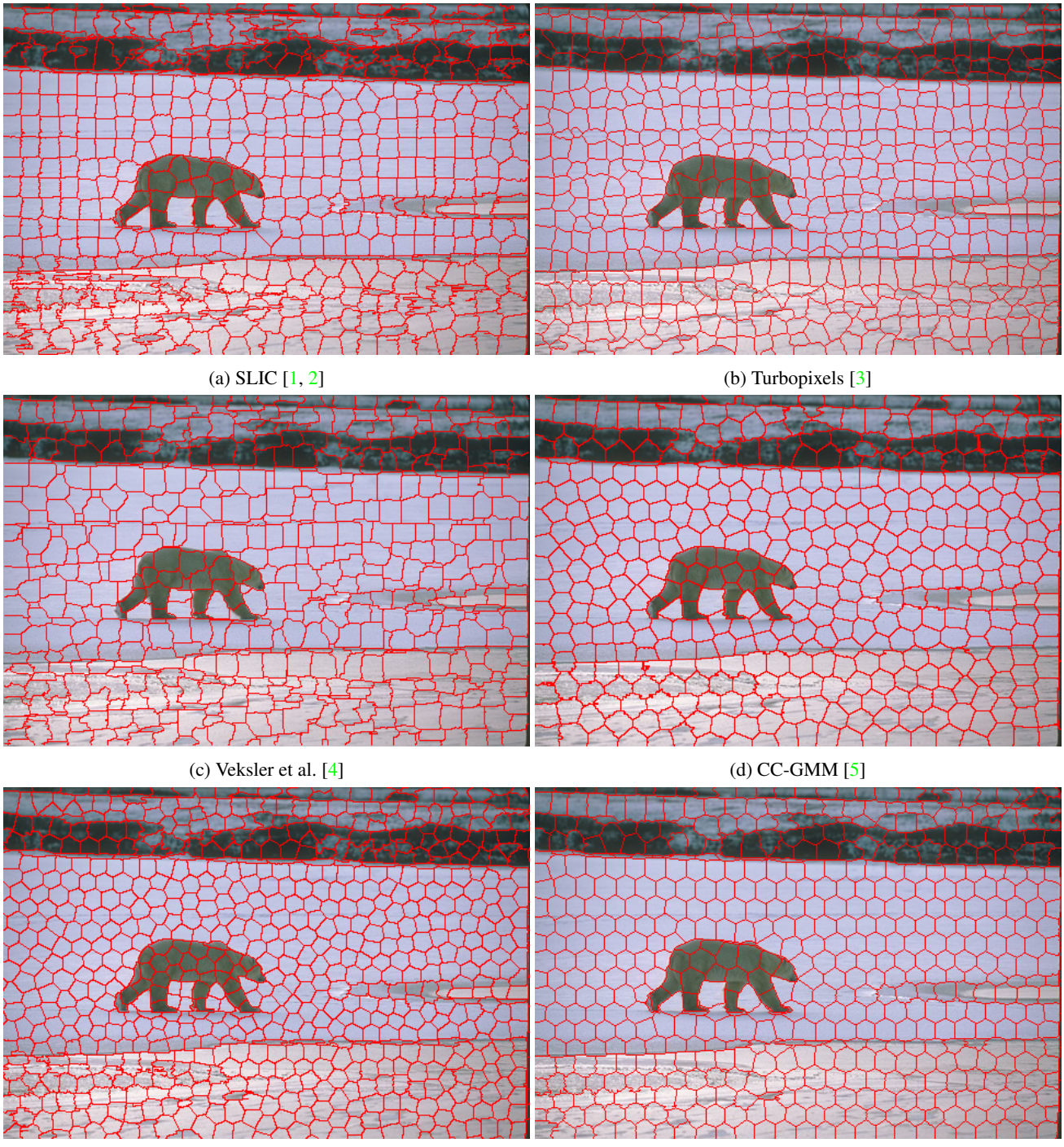
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 5: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

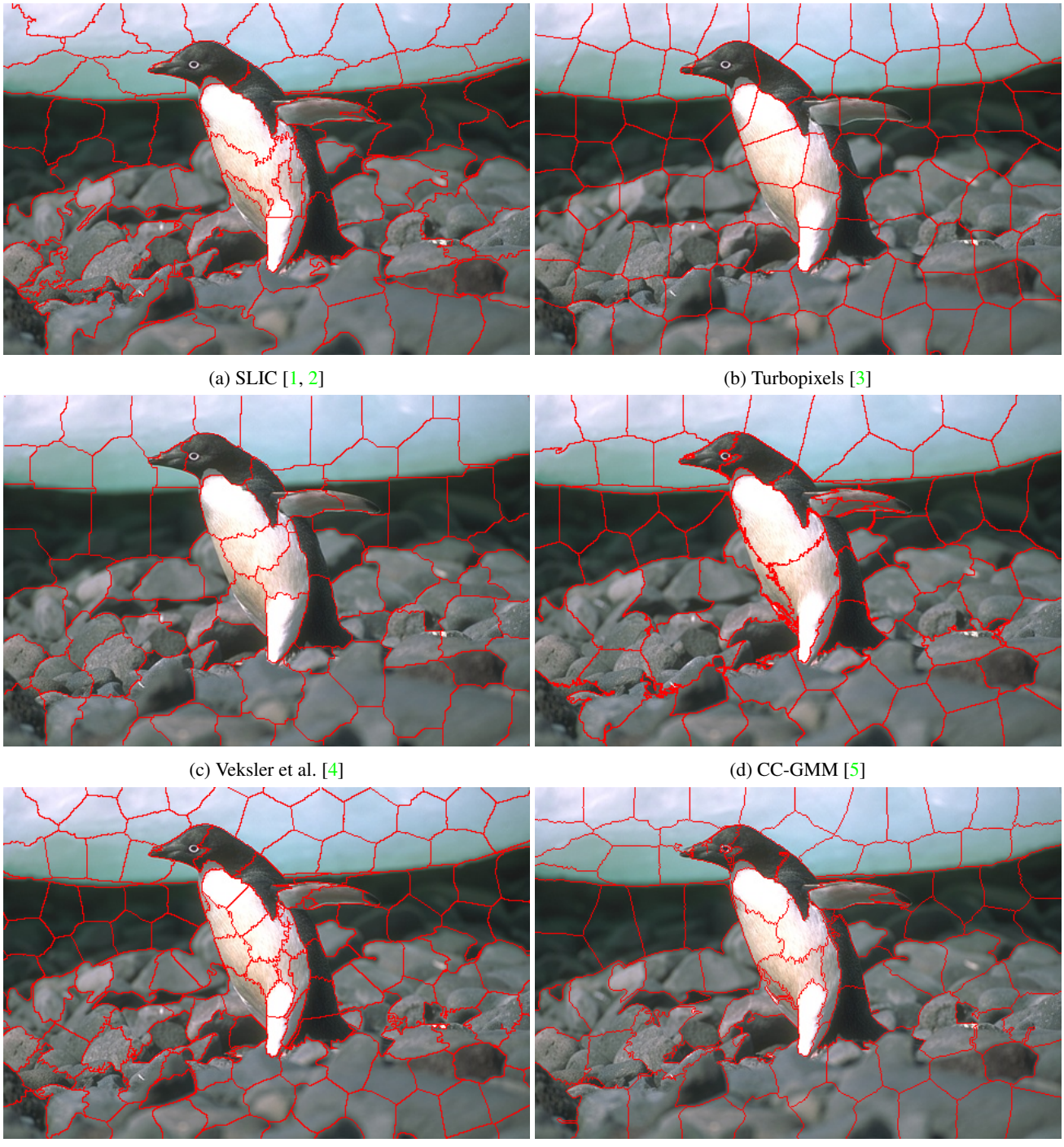
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 6: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

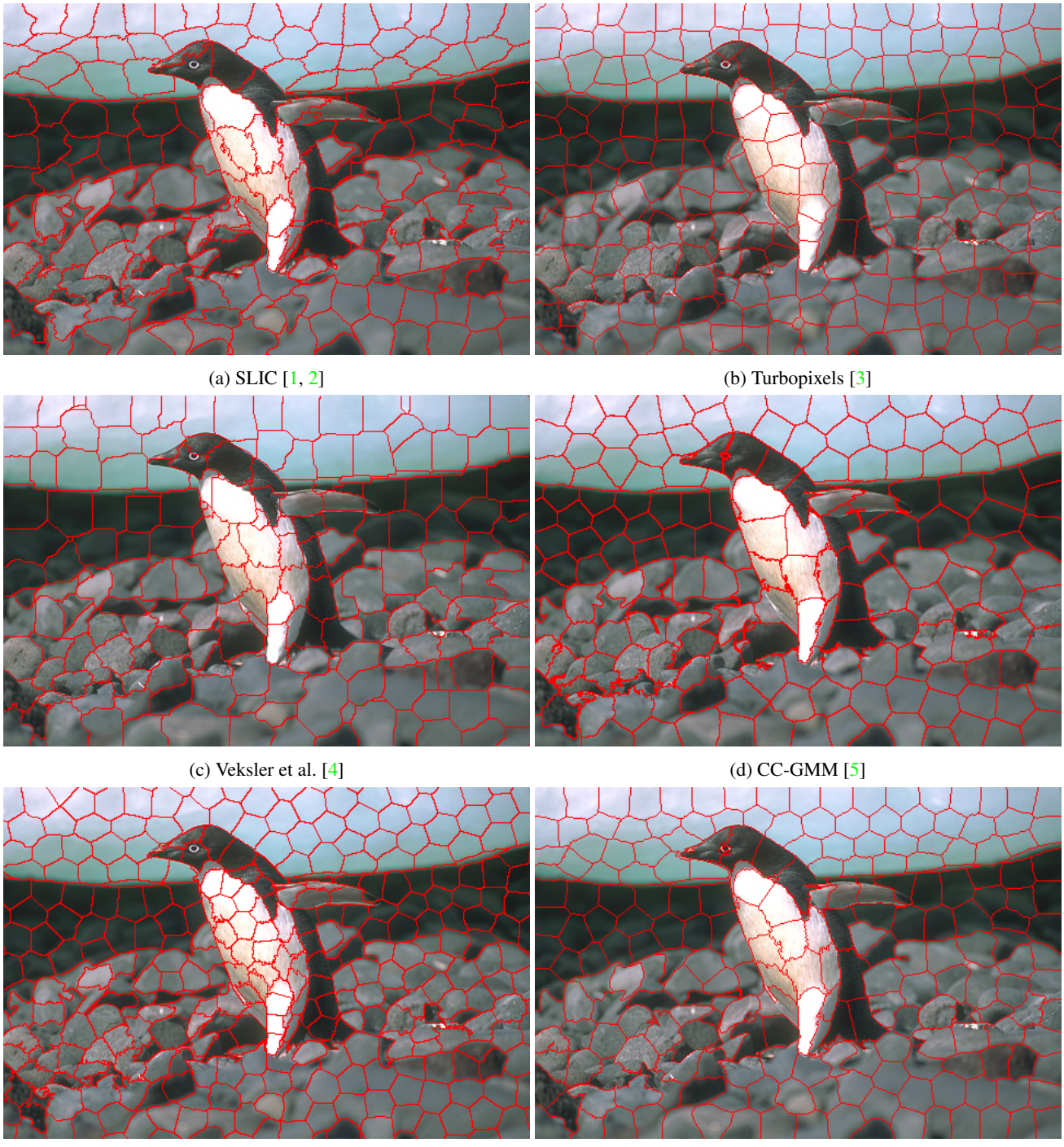
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 7: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 88$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

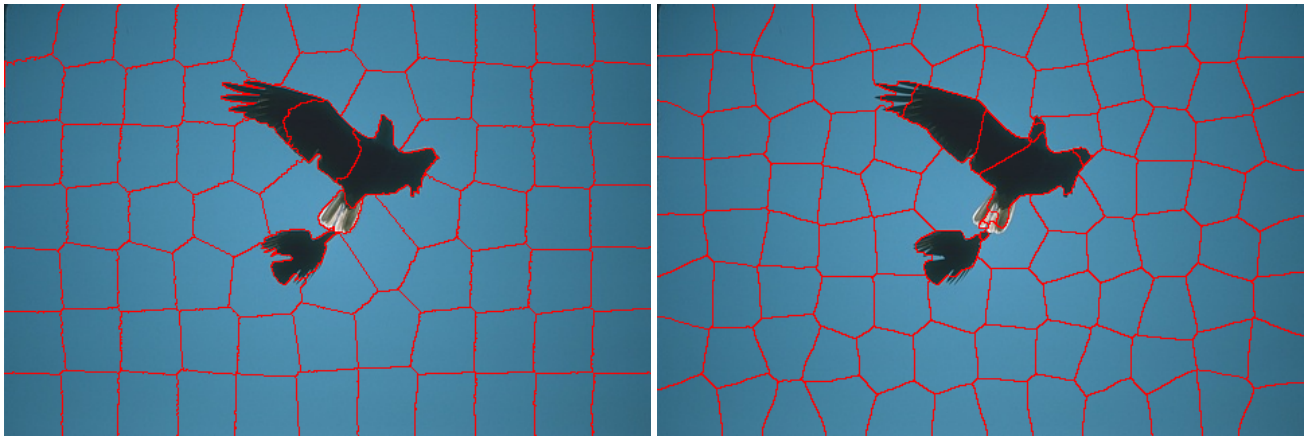
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

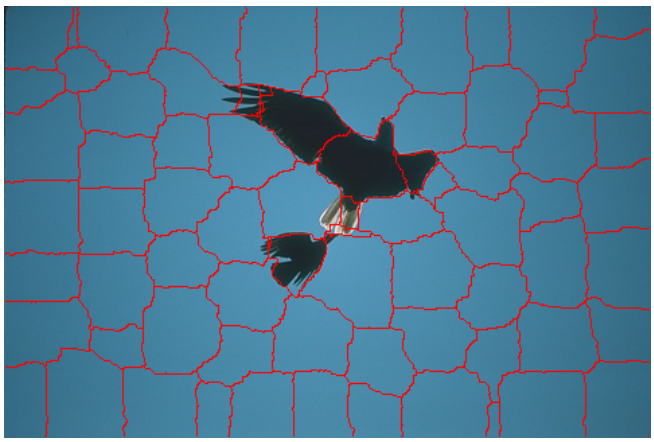
(f) Proposed Method

Fig. 8: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

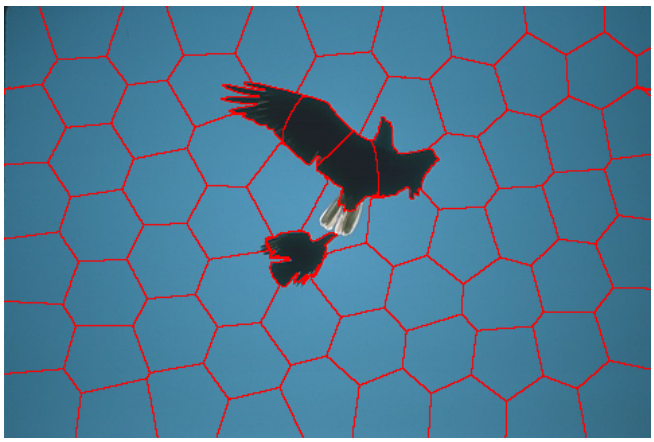


(a) SLIC [1, 2]

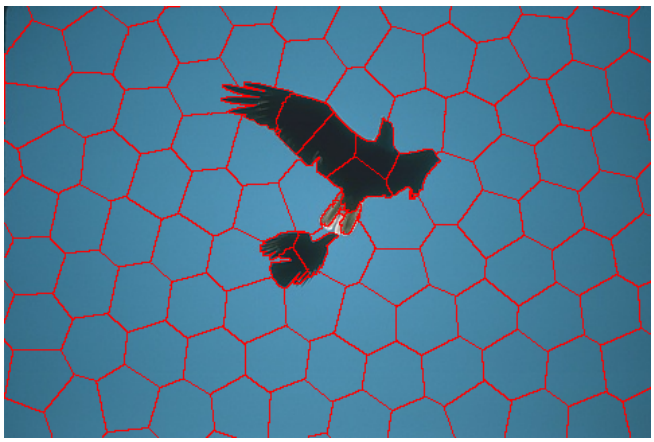
(b) Turbopixels [3]



(c) Veksler et al. [4]



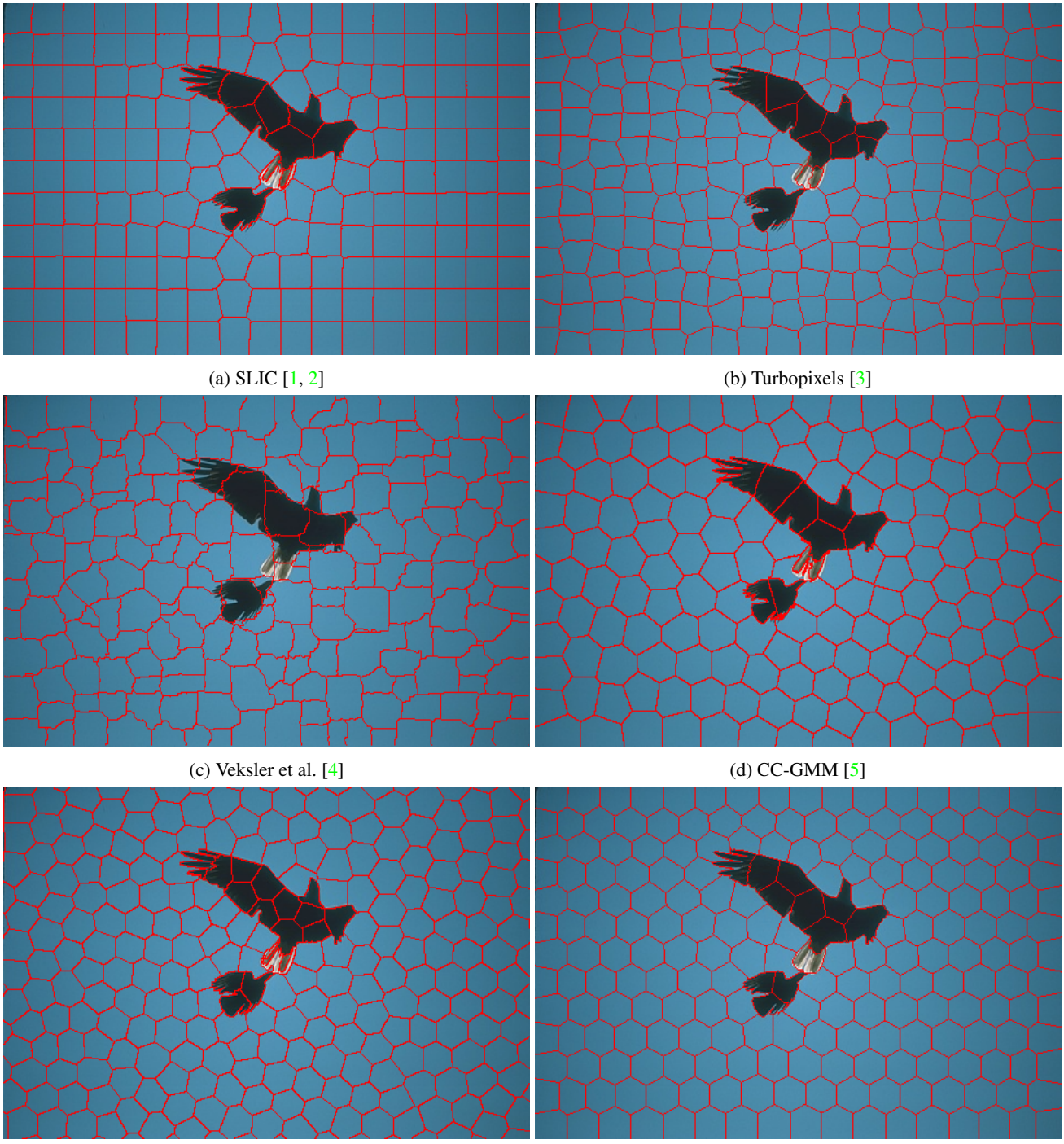
(d) CC-GMM [5]



(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 9: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 88$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

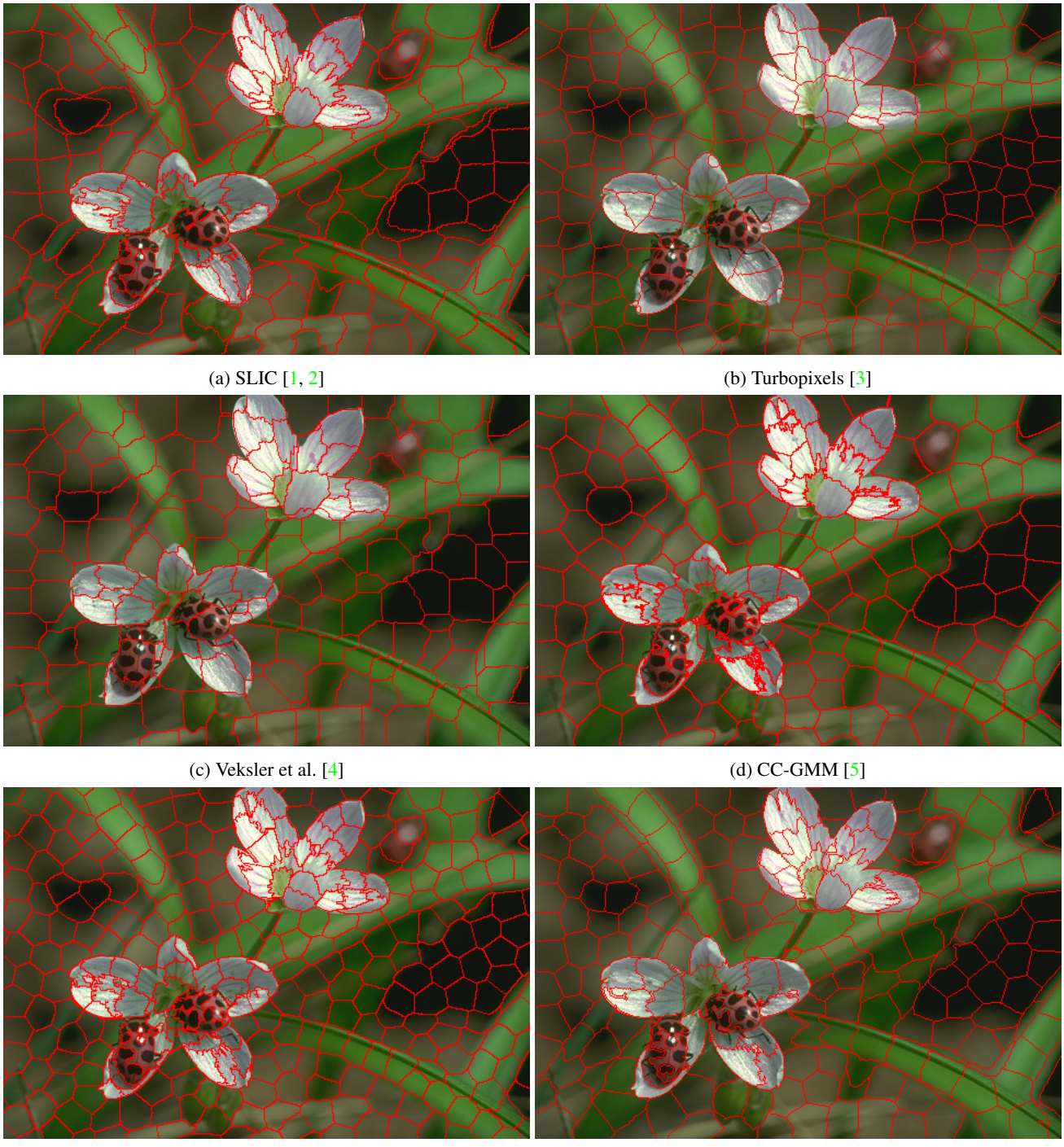
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 10: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

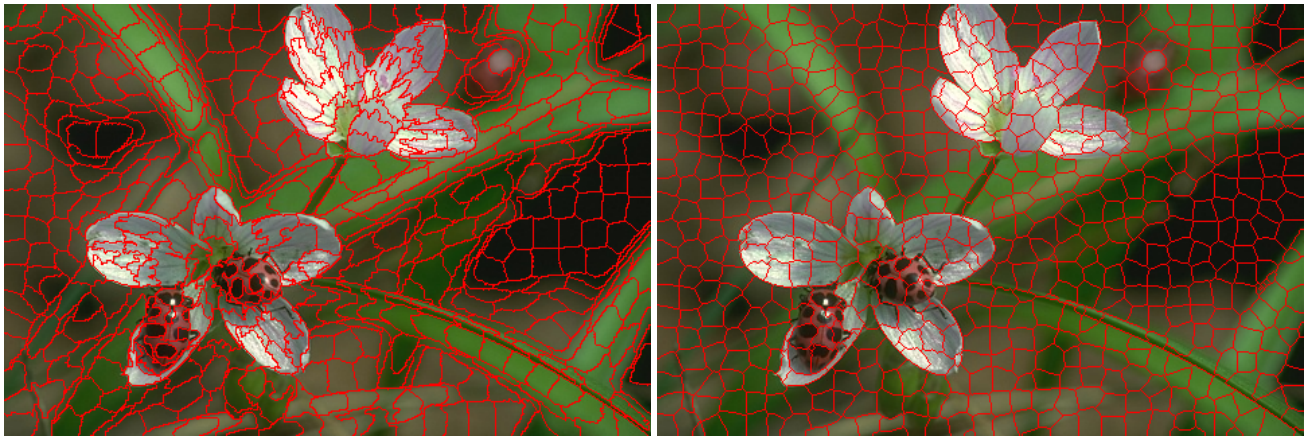
(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

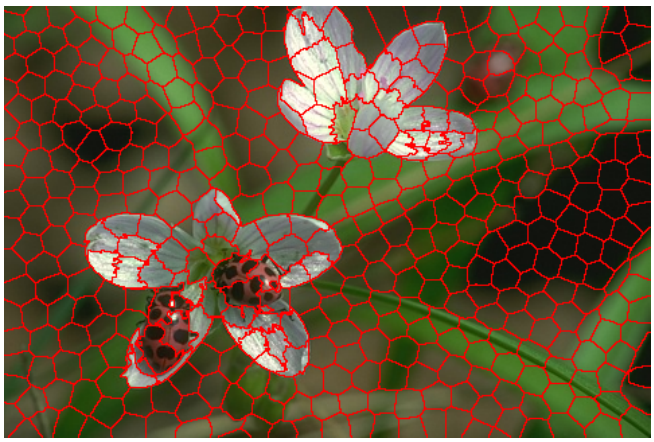
Fig. 11: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



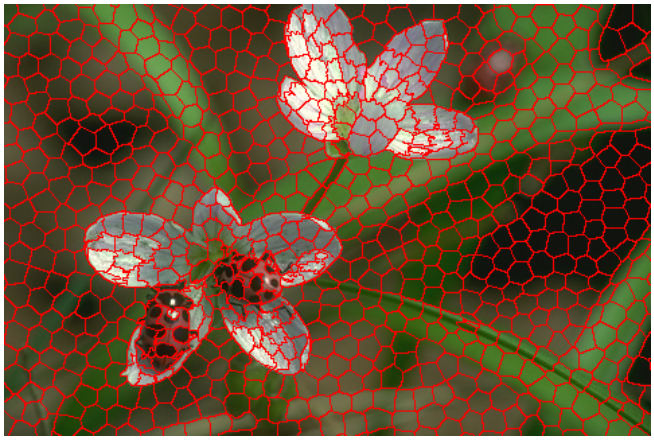
(a) SLIC [1, 2]

(b) Turbopixels [3]

(c) Veksler et al. [4]



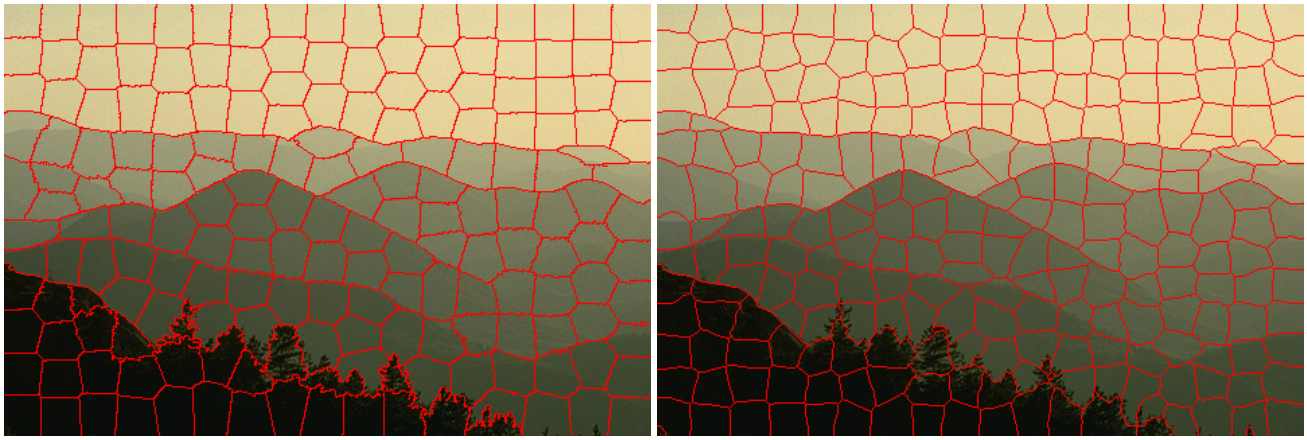
(d) CC-GMM [5]



(e) CC-GMM-SM [5]

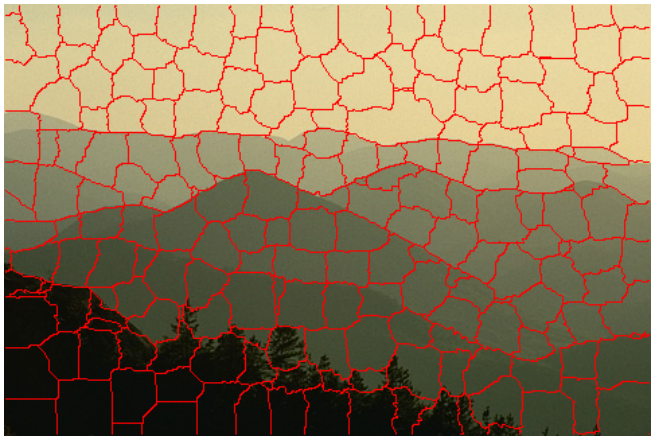
(f) Proposed Method

Fig. 12: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

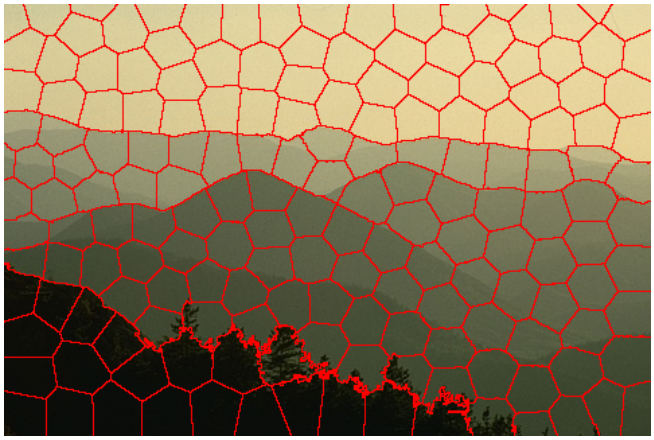


(a) SLIC [1, 2]

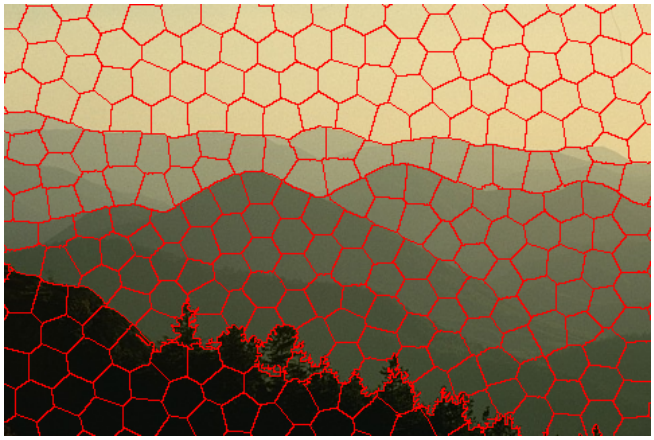
(b) Turbopixels [3]



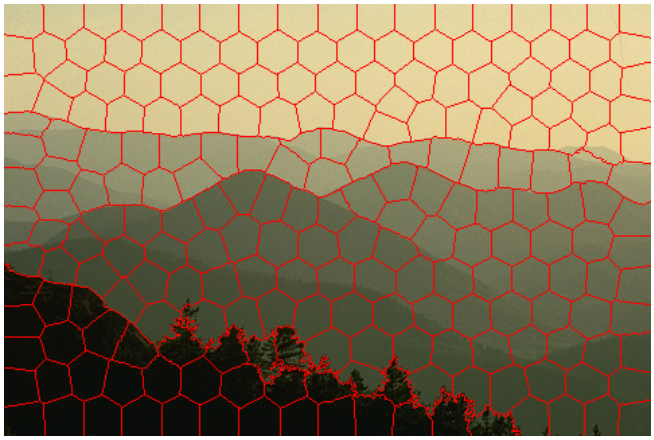
(c) Veksler et al. [4]



(d) CC-GMM [5]

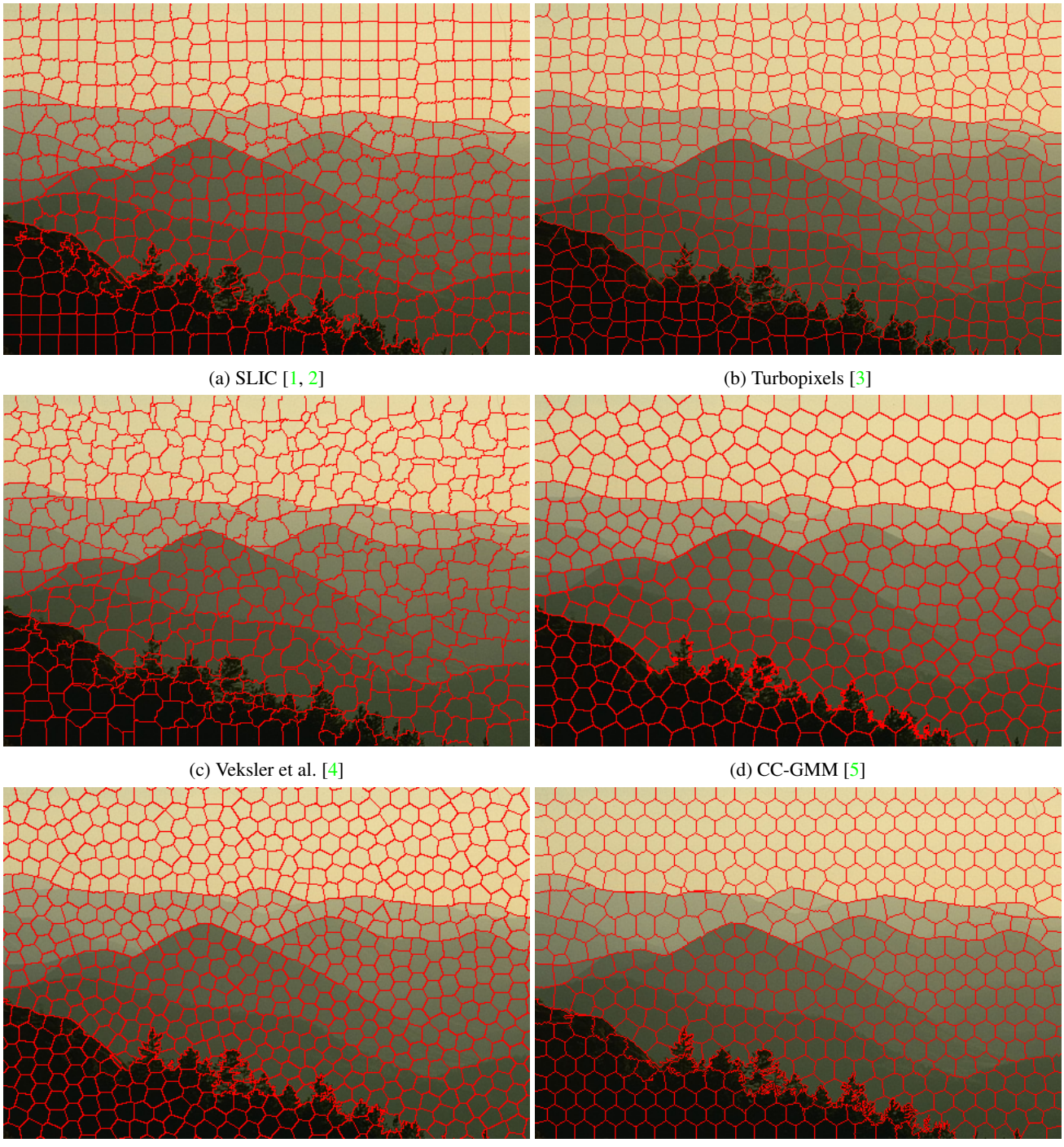


(e) CC-GMM-SM [5]



(f) Proposed Method

Fig. 13: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2]

(b) Turbopixels [3]

(c) Veksler et al. [4]

(d) CC-GMM [5]

(e) CC-GMM-SM [5]

(f) Proposed Method

Fig. 14: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

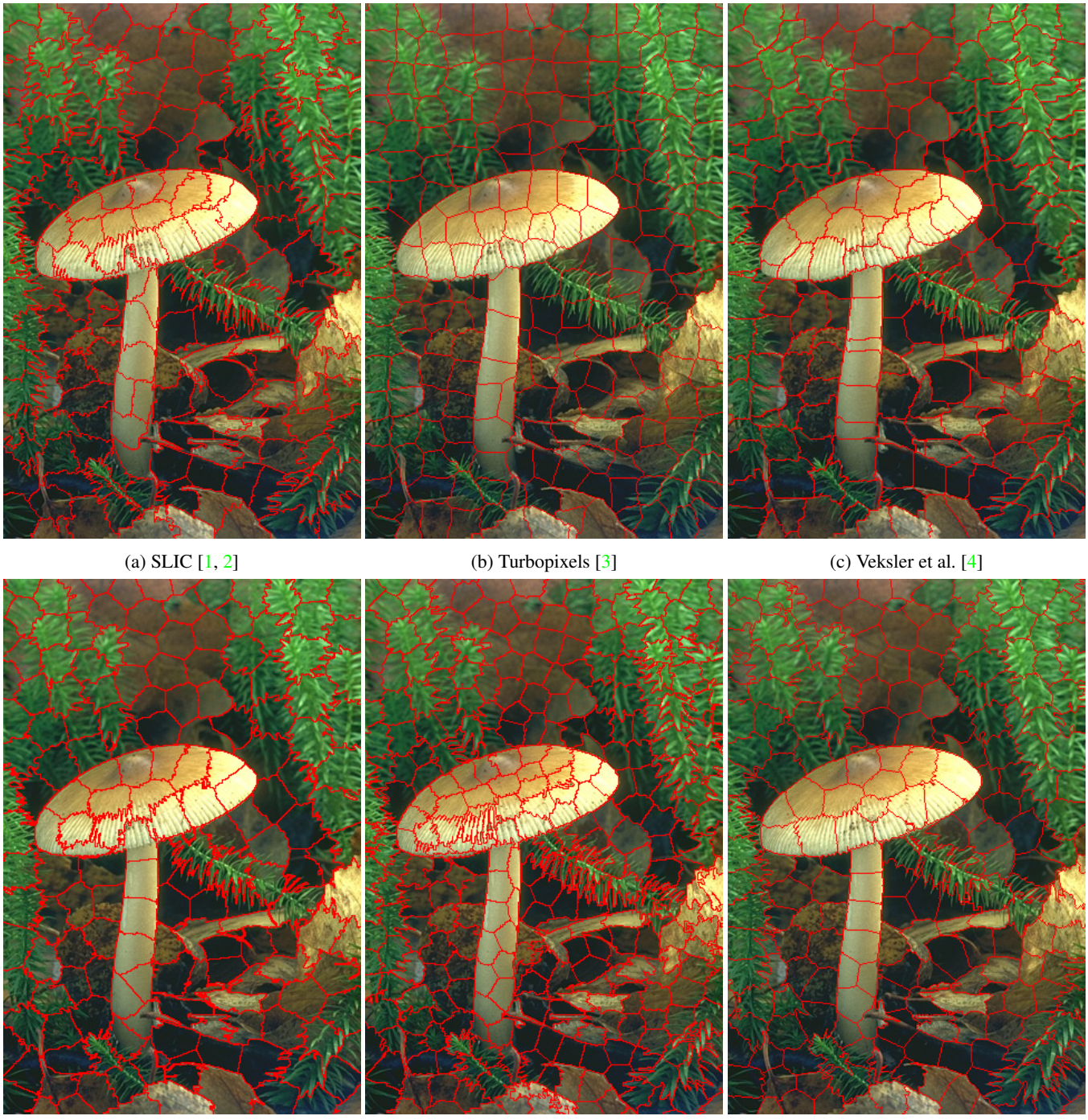
(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 15: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 88$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

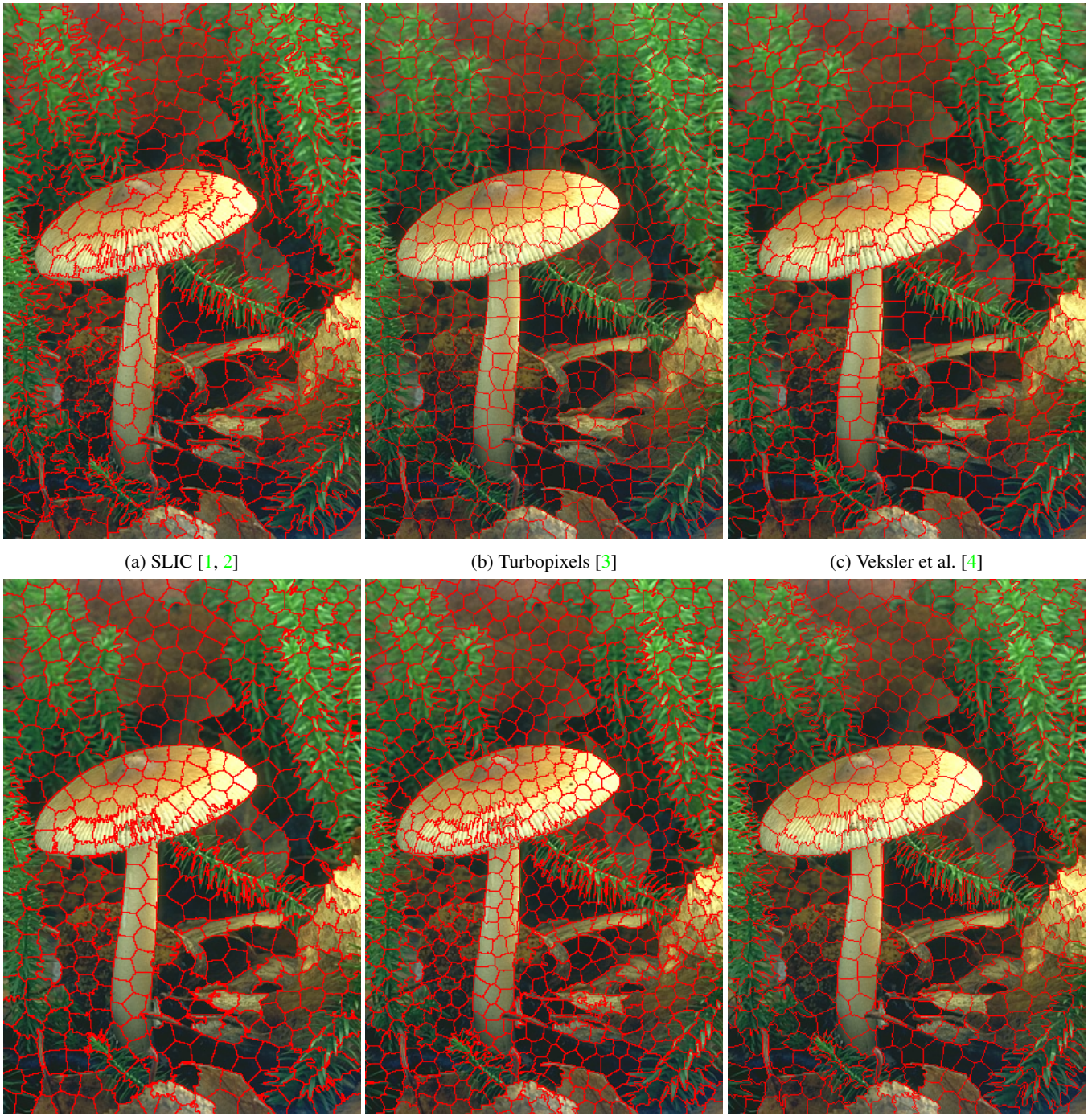
Fig. 16: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

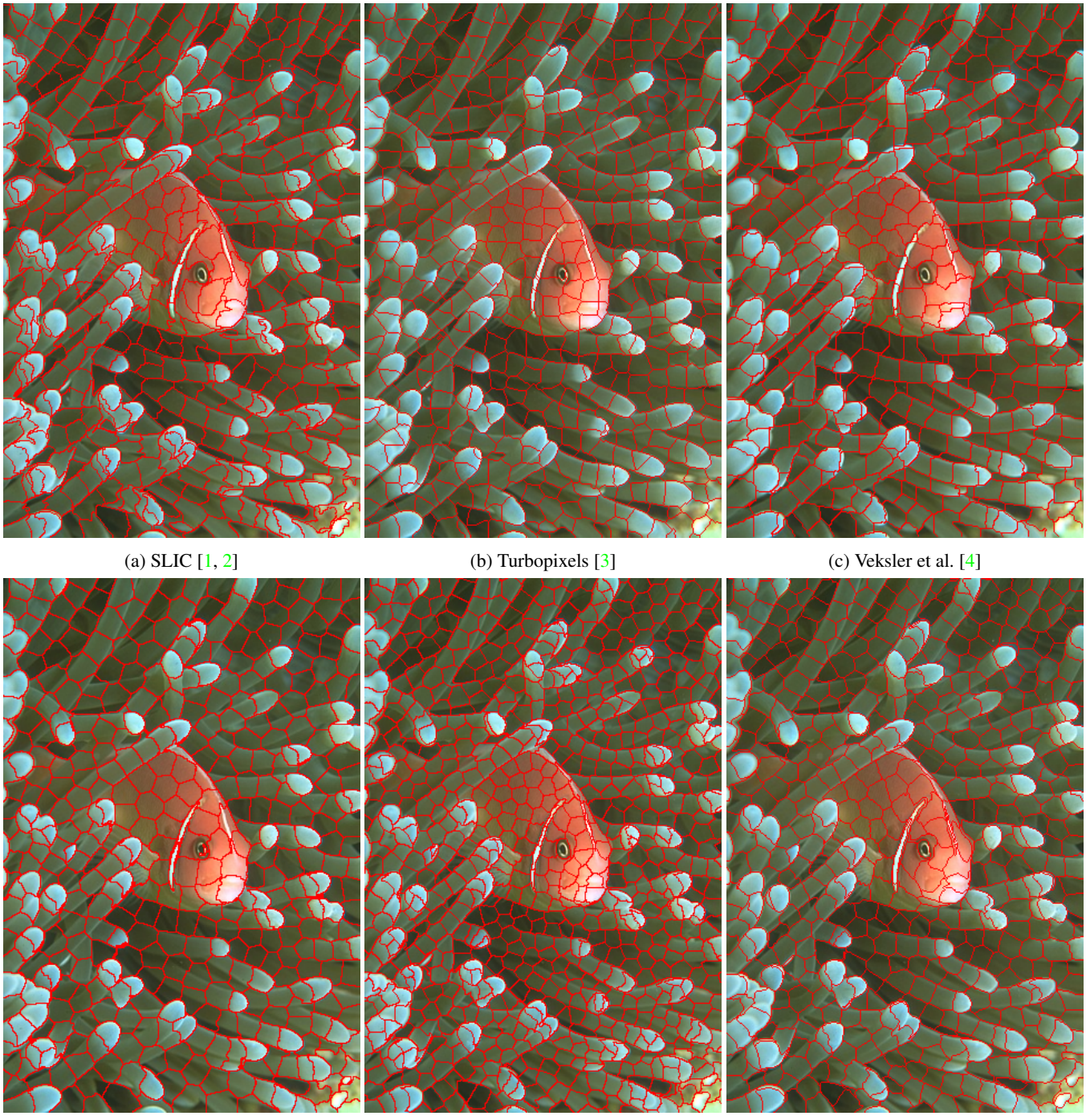
Fig. 17: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

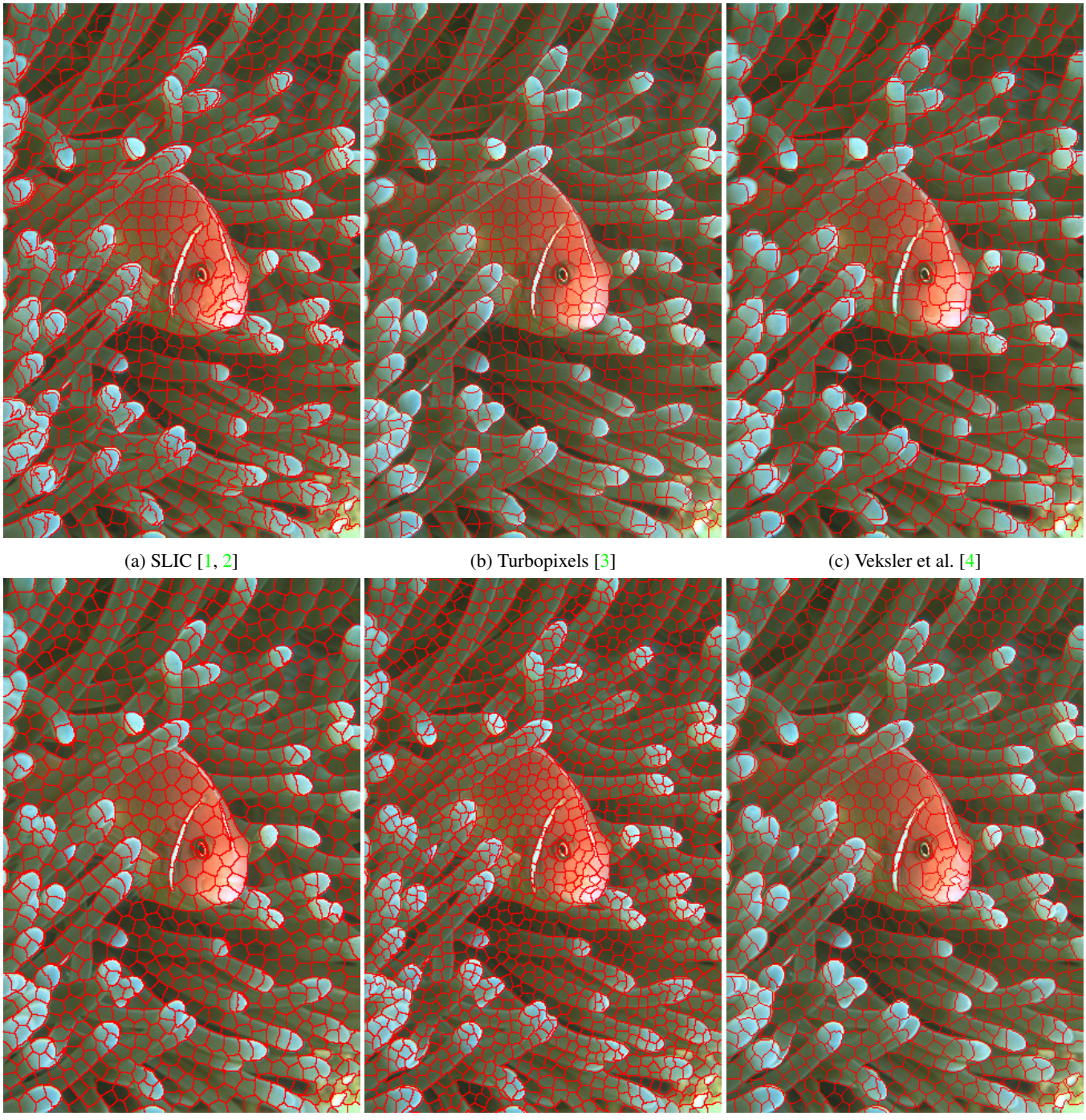
Fig. 18: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

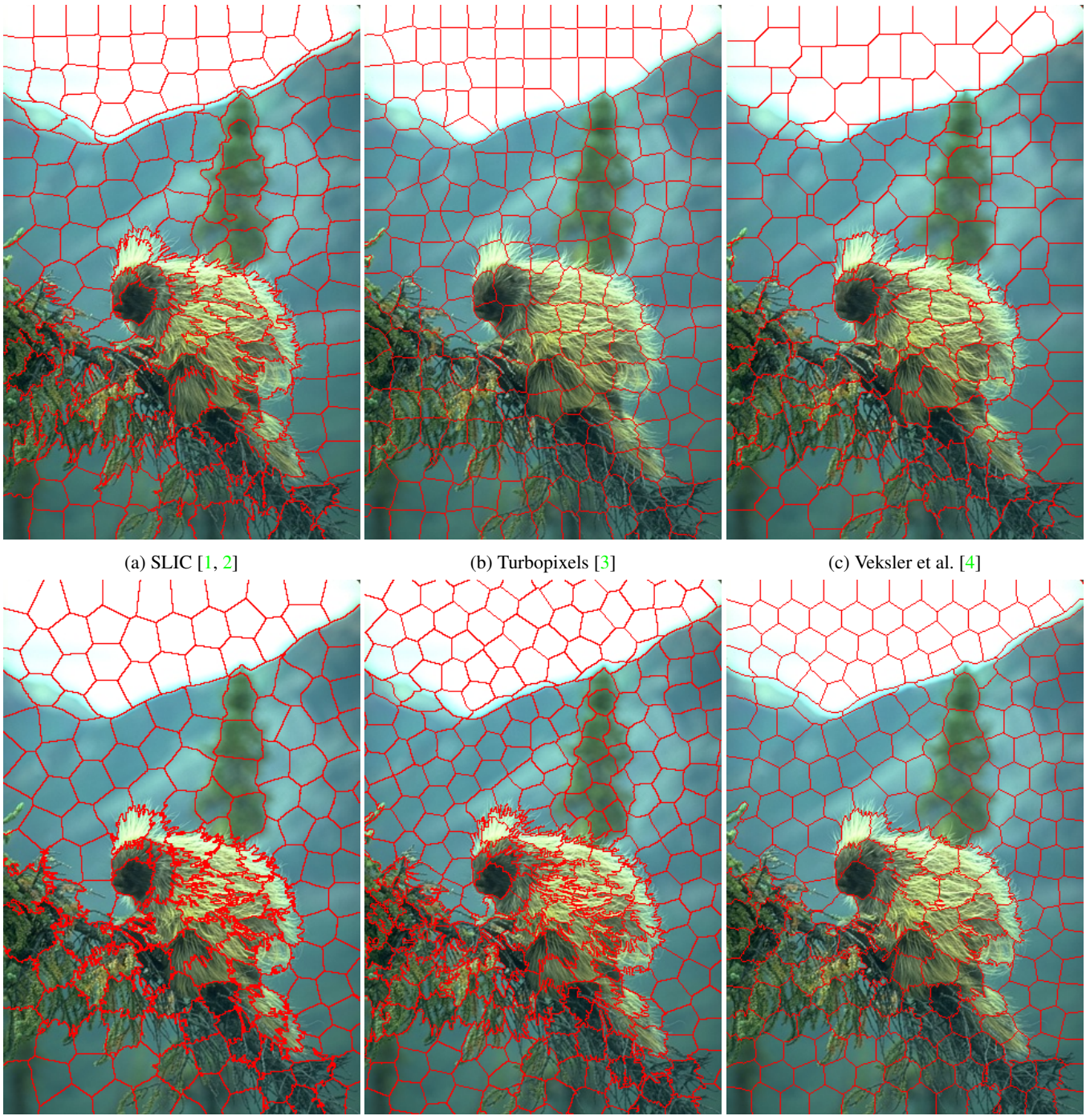
Fig. 19: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

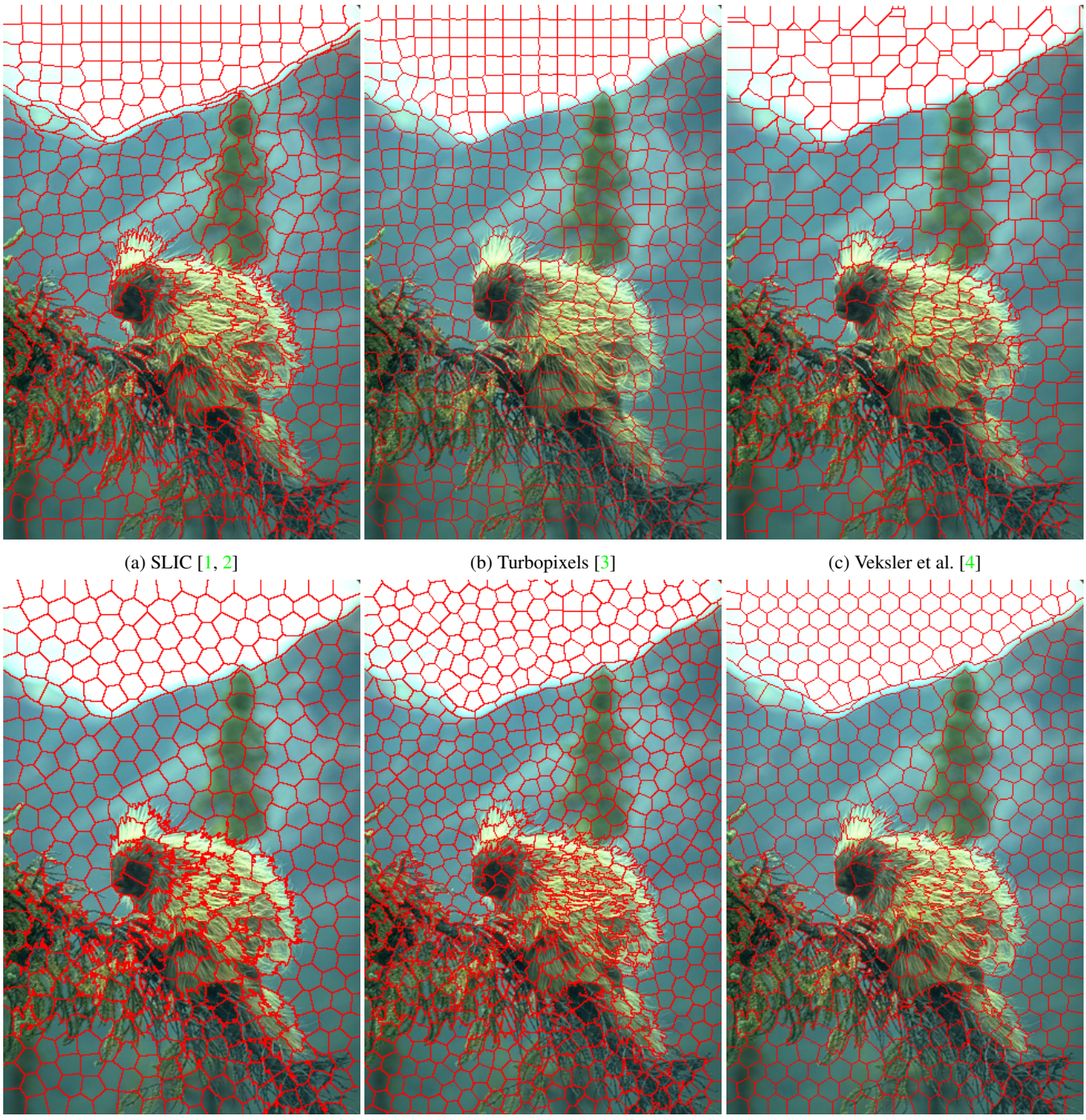
Fig. 20: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 1107$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 21: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 22: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

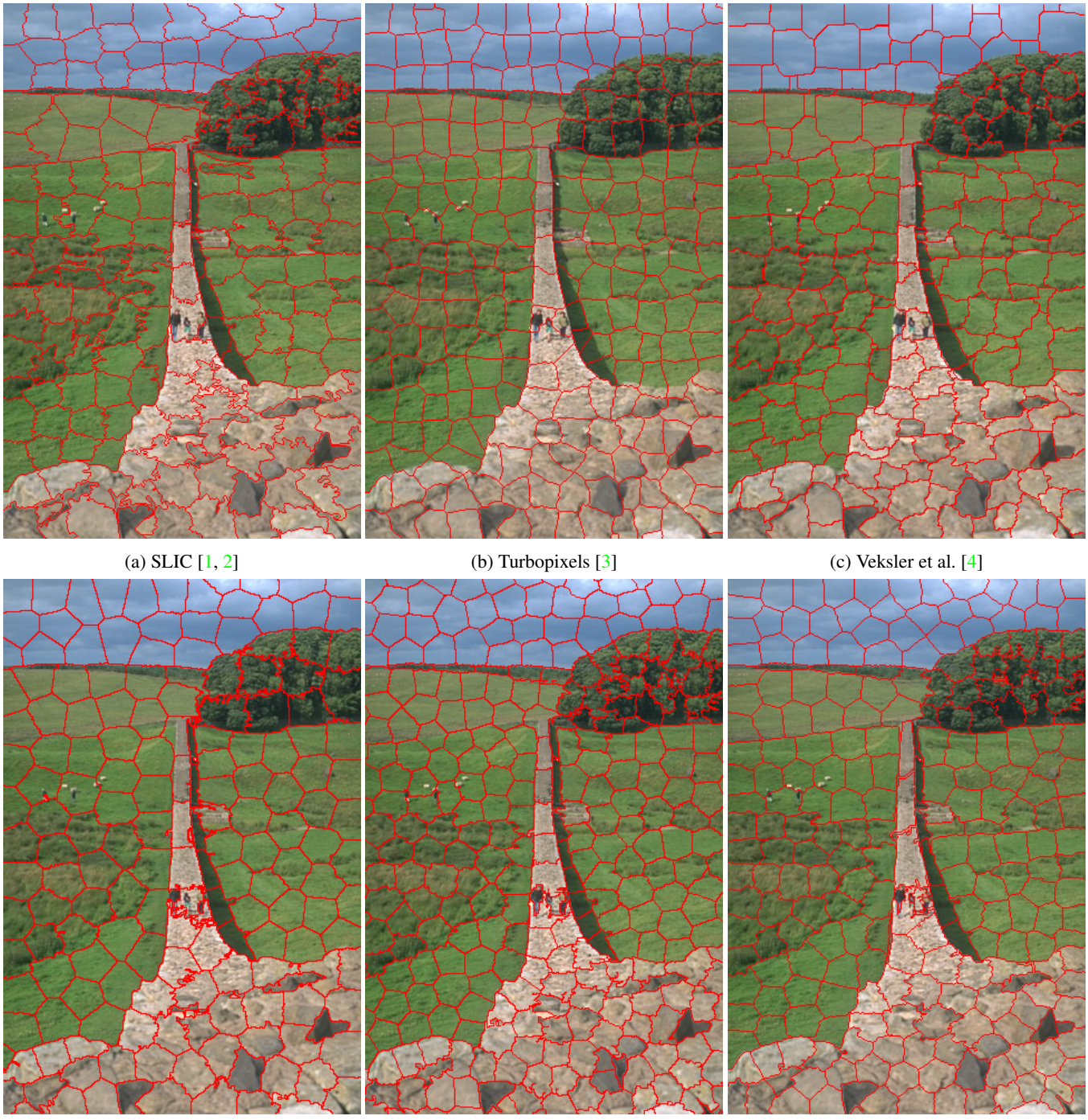
(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 23: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

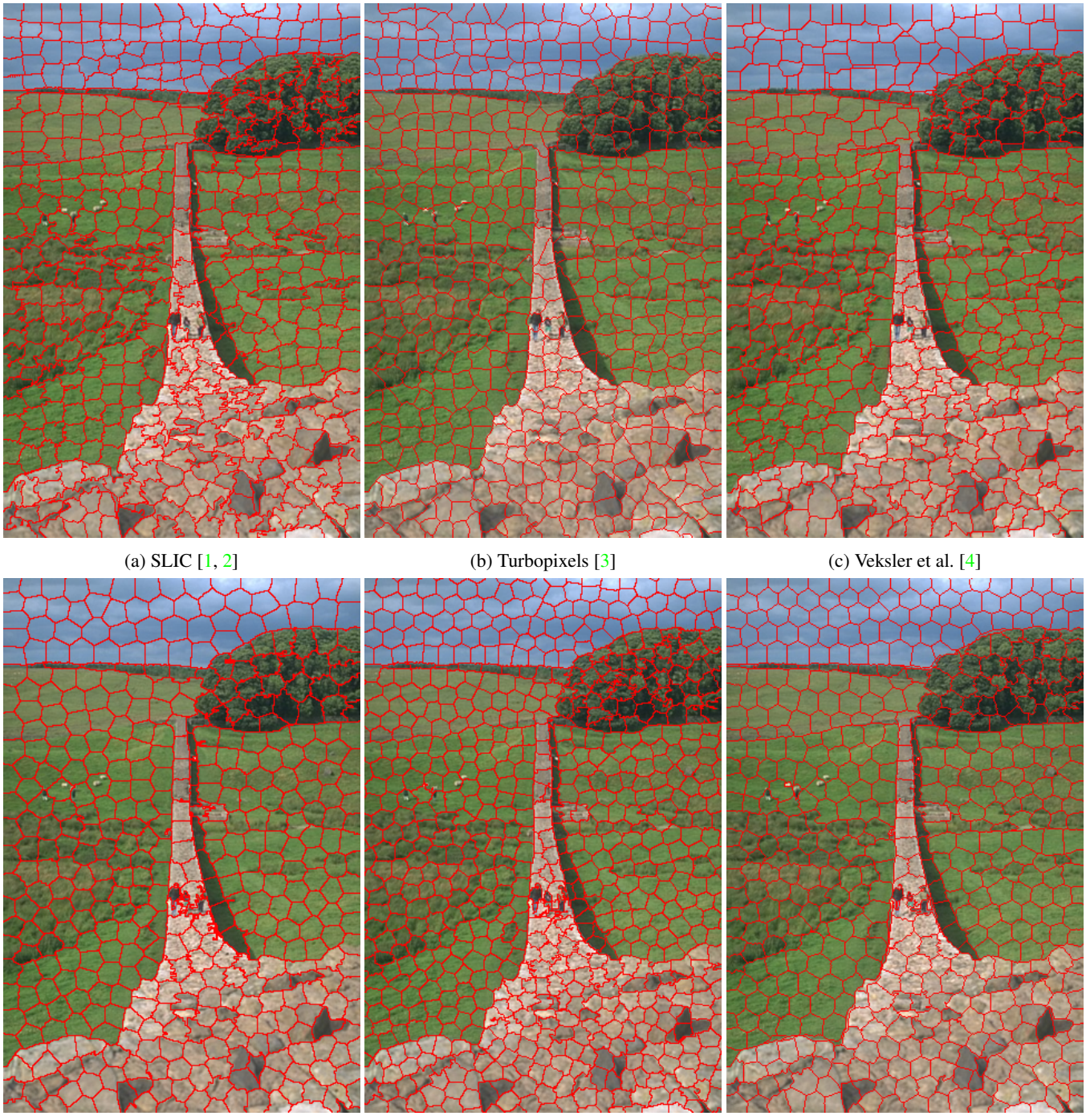
Fig. 24: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 1107$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 25: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.



(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 26: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 551$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 27: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 216$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

(a) SLIC [1, 2] (b) Turbopixels [3] (c) Veksler et al. [4]

(d) CC-GMM [5] (e) CC-GMM-SM [5] (f) Proposed Method

Fig. 28: A visual comparison. Image is taken from the BSDS500 dataset [6]. Expected $K = 1107$. The actual value of K may vary a little across the different methods. E.g., in CC-GMM, CC-GMM-SM and the proposed method, superpixels can "die off" during inference if their weights become zero. Additionally, CC-GMM-SM also uses a few steps of splits and merges.

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